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Roll no. :4023

Urn: 1022101007

Subject: Data Analytics with R

Experiment No. 04

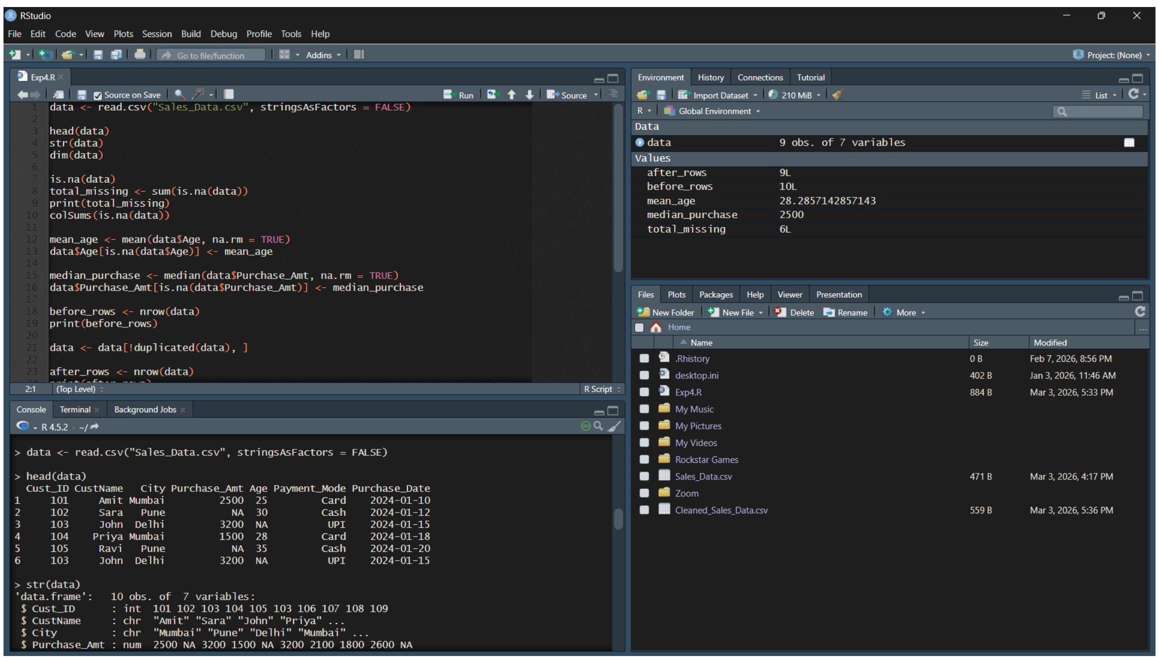
To implement a program to read data from a CSV file and clean it using R

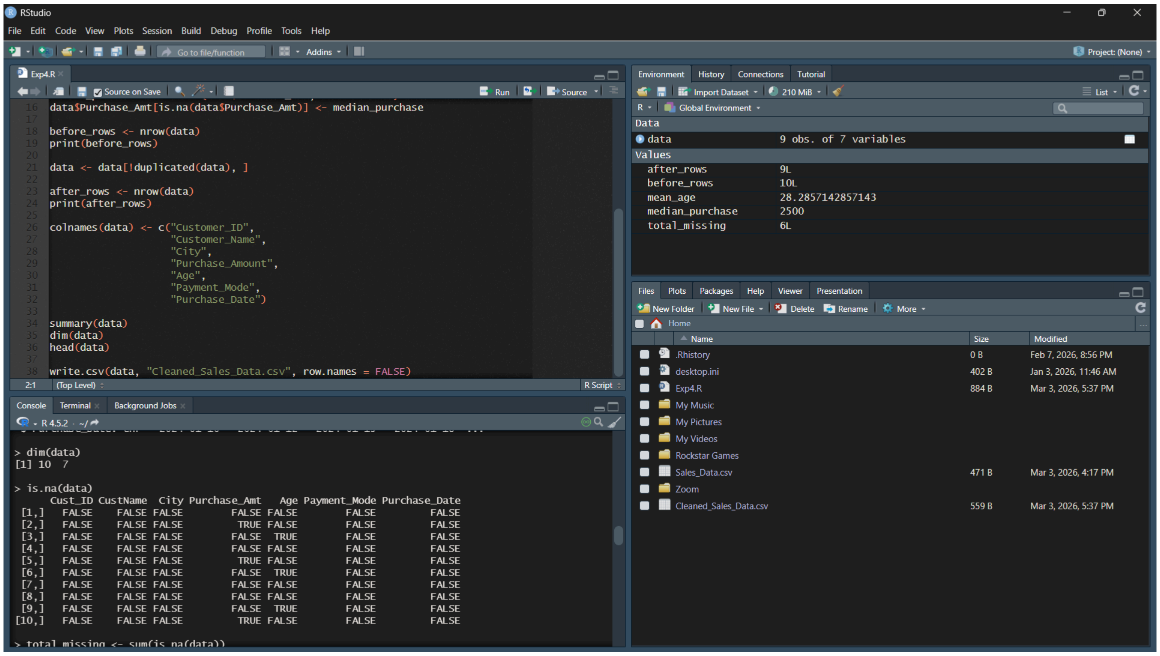
Title: To implement a program to read data from a CSV file and perform data cleaning using R (Handling missing values, removing duplicates and renaming columns)

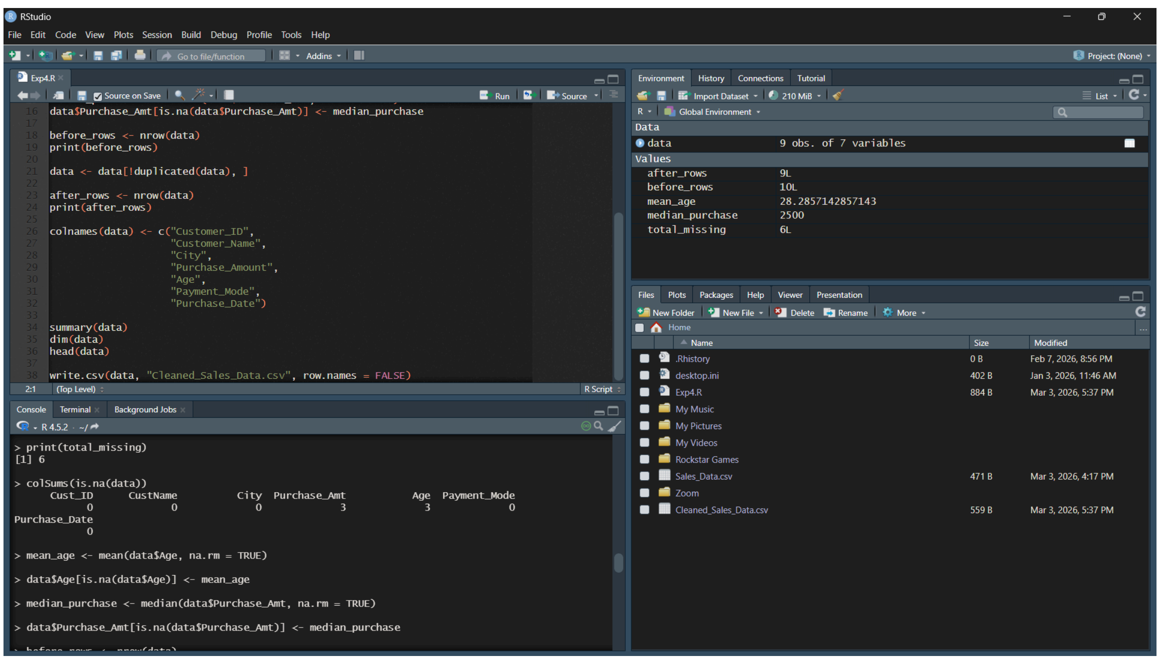
Aim:

To read data from a CSV file into R and perform essential data cleaning operations such as handling missing values, removing duplicate records and renaming columns to prepare the dataset for analysis

Code / Outputs:




RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Exp4.R
data$Purchase_Amt[is.na(data$Purchase_Amt)] <- median_purchase
before_rows <- nrow(data)
print(before_rows)
data <- data[!duplicated(data),]
after_rows <- nrow(data)
print(after_rows)
colnames(data) <- c("Customer_ID",
"Customer_Name",
"City",
"Purchase_Amount",
"Age",
"Payment_Mode",
"Purchase_Date")
summary(data)
dim(data)
head(data)
write.csv(data, "Cleaned_Sales_Data.csv", row.names = FALSE)
Console Terminal Background Jobs
> dim(data)
[1] 10 7
> is.na(data)
Cust_ID CustName City Purchase_Amt Age Payment_Mode Purchase_Date
[1,] FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[2,] FALSE FALSE FALSE TRUE FALSE FALSE FALSE
[3,] FALSE FALSE FALSE FALSE TRUE FALSE FALSE
[4,] FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[5,] FALSE FALSE FALSE TRUE FALSE FALSE FALSE
[6,] FALSE FALSE FALSE FALSE TRUE FALSE FALSE
[7,] FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[8,] FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[9,] FALSE FALSE FALSE FALSE TRUE FALSE FALSE
[10,] FALSE FALSE FALSE TRUE FALSE FALSE FALSE
> total_missing <- sum(is.na(data))
Environment History Connections Tutorial
Global Environment
Data
data 9 obs. of 7 variables
Values
after_rows 9L
before_rows 10L
mean_age 28.2857142857143
median_purchase 2500
total_missing 6L
Files Plots Packages Help Viewer Presentation
Home
Name Size Modified
.Rhistory 0 B Feb 7, 2026, 8:56 PM
desktop.ini 402 B Jan 3, 2026, 11:46 AM
Exp4.R 884 B Mar 3, 2026, 5:37 PM
My Music
My Pictures
My Videos
Rockstar Games
Sales_Data.csv 471 B Mar 3, 2026, 4:17 PM
Zoom
Cleaned_Sales_Data.csv 559 B Mar 3, 2026, 5:37 PM


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Exp4.R
data$Purchase_Amt[is.na(data$Purchase_Amt)] <- median_purchase
before_rows <- nrow(data)
print(before_rows)
data <- data[!duplicated(data),]
after_rows <- nrow(data)
print(after_rows)
colnames(data) <- c("Customer_ID",
"Customer_Name",
"City",
"Purchase_Amount",
"Age",
"Payment_Mode",
"Purchase_Date")
summary(data)
dim(data)
head(data)
write.csv(data, "Cleaned_Sales_Data.csv", row.names = FALSE)
Console Terminal Background Jobs
> print(total_missing)
[1] 6
> colSums(is.na(data))
Cust_ID CustName City Purchase_Amt Age Payment_Mode
0 0 0 3 3 0
Purchase_Date
0
> mean_age <- mean(data$Age, na.rm = TRUE)
> data$Age[is.na(data$Age)] <- mean_age
> median_purchase <- median(data$Purchase_Amt, na.rm = TRUE)
> data$Purchase_Amt[is.na(data$Purchase_Amt)] <- median_purchase
Environment History Connections Tutorial
Global Environment
Data
data 9 obs. of 7 variables
Values
after_rows 9L
before_rows 10L
mean_age 28.2857142857143
median_purchase 2500
total_missing 6L
Files Plots Packages Help Viewer Presentation
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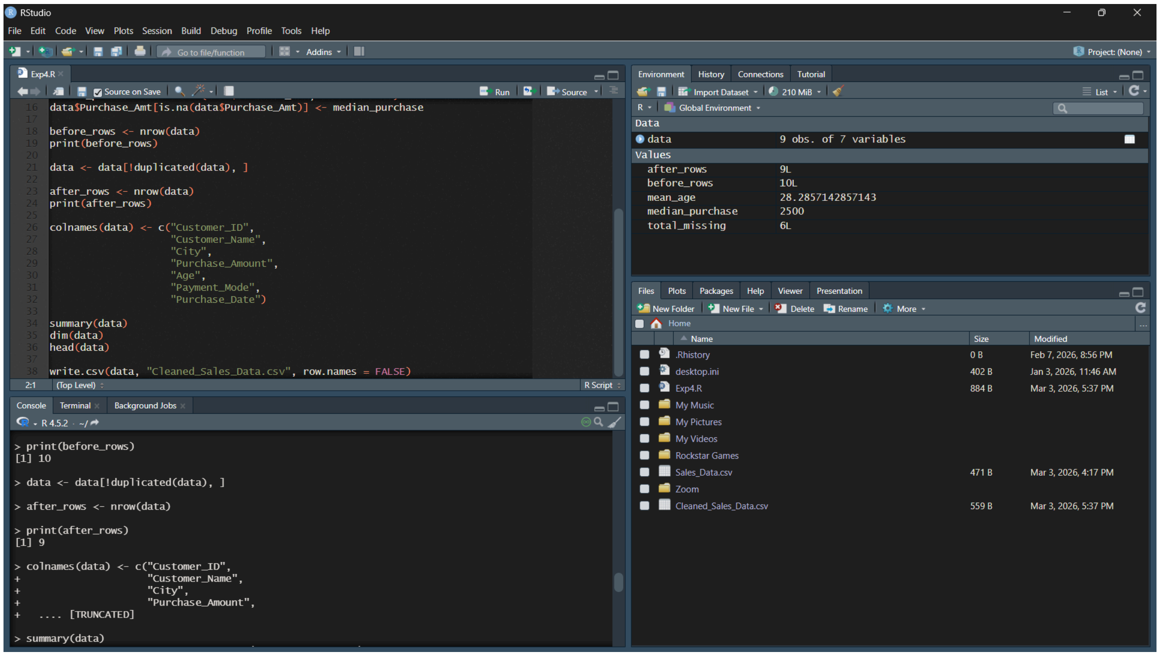
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Exp4.R
Source on Save
Run
Source
data$Purchase_Amt[is.na(data$Purchase_Amt)] <- median_purchase
before_rows <- nrow(data)
print(before_rows)
data <- data[!duplicated(data),]
after_rows <- nrow(data)
print(after_rows)
colnames(data) <- c("Customer_ID",
"Customer_Name",
"City",
"Purchase_Amount",
"Age",
"Payment_Mode",
"Purchase_Date")
summary(data)
dim(data)
head(data)
write.csv(data, "Cleaned_Sales_Data.csv", row.names = FALSE)
2:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.5.2 · ~/
> print(before_rows)
[1] 10
> data <- data[!duplicated(data),]
> after_rows <- nrow(data)
> print(after_rows)
[1] 9
> colnames(data) <- c("Customer_ID",
+ "Customer_Name",
+ "City",
+ "Purchase_Amount",
+ .... [TRUNCATED]
> summary(data)
Environment
History
Connections
Tutorial
Import Dataset
210 MiB
List
R
Global Environment
Data
data 9 obs. of 7 variables
Values
after_rows 9L
before_rows 10L
mean_age 28.2857142857143
median_purchase 2500
total_missing 6L
Files
Plots
Packages
Help
Viewer
Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
.Rhistory 0 B Feb 7, 2026, 8:56 PM
desktop.ini 402 B Jan 3, 2026, 11:46 AM
Exp4.R 884 B Mar 3, 2026, 5:37 PM
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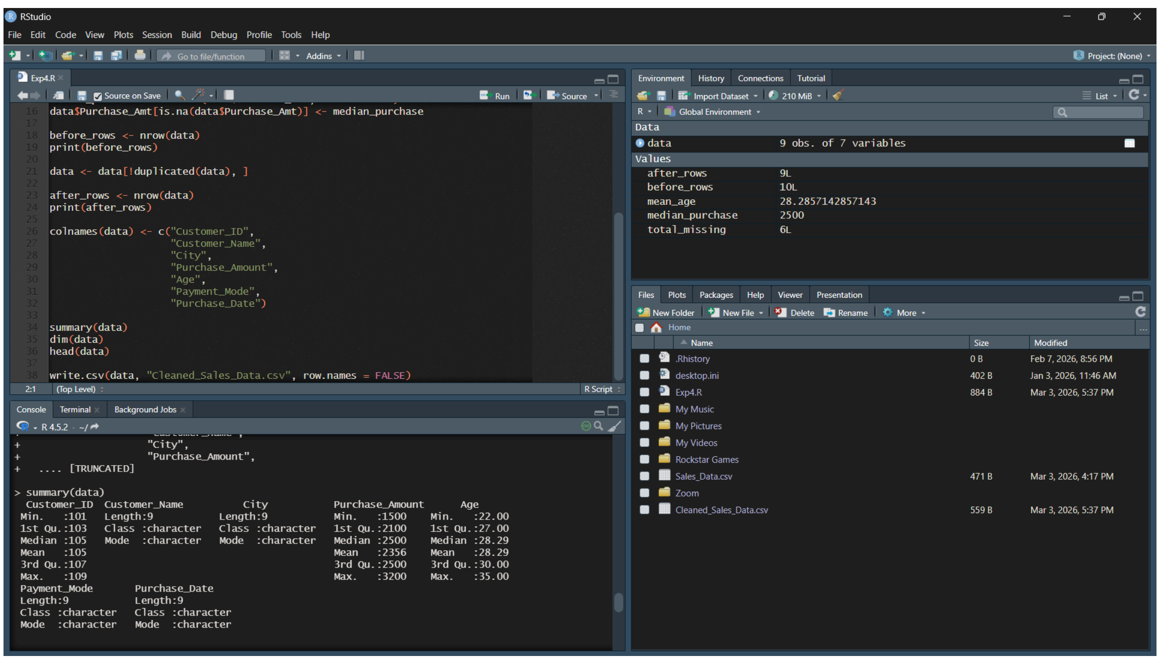
Zoom
Cleaned_Sales_Data.csv 559 B Mar 3, 2026, 5:37 PM


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Exp4.R
Source on Save
Run
Source
data$Purchase_Amt[is.na(data$Purchase_Amt)] <- median_purchase
before_rows <- nrow(data)
print(before_rows)
data <- data[!duplicated(data),]
after_rows <- nrow(data)
print(after_rows)
colnames(data) <- c("Customer_ID",
"Customer_Name",
"City",
"Purchase_Amount",
"Age",
"Payment_Mode",
"Purchase_Date")
summary(data)
dim(data)
head(data)
write.csv(data, "Cleaned_Sales_Data.csv", row.names = FALSE)
2:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.5.2 · ~/
+ "City",
+ "Purchase_Amount",
+ .... [TRUNCATED]
> summary(data)
Customer_ID Customer_Name City Purchase_Amount Age
Min. :101 Length:9 Length:9 Min. :1500 Min. :22.00
1st Qu.:103 Class :character Class :character 1st Qu.:2100 1st Qu.:27.00
Median :105 Mode :character Mode :character Median :2500 Median :28.29
Mean :105 Mean :2356 Mean :28.29
3rd Qu.:107 3rd Qu.:2500 3rd Qu.:30.00
Max. :109 Max. :3200 Max. :35.00
Payment_Mode Purchase_Date
Length:9 Length:9
Class :character Class :character
Mode :character Mode :character
Environment
History
Connections
Tutorial
Import Dataset
210 MiB
List
R
Global Environment
Data
data 9 obs. of 7 variables
Values
after_rows 9L
before_rows 10L
mean_age 28.2857142857143
median_purchase 2500
total_missing 6L
Files
Plots
Packages
Help
Viewer
Presentation
New Folder
New File
Delete
Rename
More
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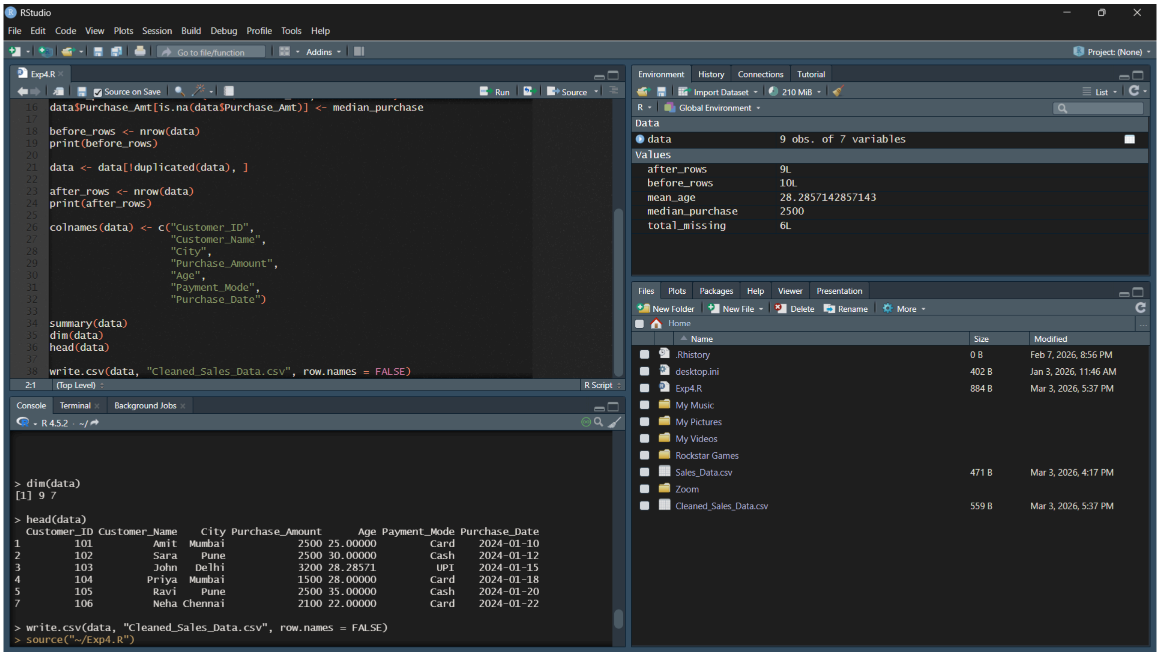

Conclusion:

The experiment demonstrated how to import and clean datasets using R. students learned to identify missing values, remove duplicates and rename columns effectively. Data cleaning ensures improved data quality and reliability, forming a crucial preprocessing step before performing analytical or predictive tasks